Capito: learn money by growing a practice portfolio

Vanguard India brief: a digital financial-literacy experience for Grade 11–12 students in underserved Hyderabad communities.

1. Problem and our reading of the brief

Students meet money decisions before anyone teaches them: festival sales, "no-cost" EMIs, instant-loan apps, UPI refund scams, first stipends. The brief asks for something students *want* to use, starting with about 1,500 learners and designed to scale, that lets them make choices, see consequences and try again. It must be simple, relatable, game-based and language-ready, work with NGOs, educate without recommending products, label simulations as assumptions, and respect privacy.

What makes it hard: learners are minors on shared, low-end phones with patchy data; mistakes must feel real but cost nothing; financial words must stay recognisable across English, Hindi and Telugu; and the "never recommend" guardrail has to be built in, not just stated.

2. The solution

Learning is growing a practice portfolio. Each of the 29 lessons is one game-month:

Gut-check decision → **5–8 cards** → **Built-in tool** → **3-question quiz** → **Practice money** → **Plan the Month** → **Life events**

- **Decision first:** every lesson opens with a choice (buy the sneakers, save it all, or split) and shows where it leads before any teaching. Nothing is spent.
- **Plain words first:** one idea per card; each money word opens a Word Bank card in three layers: say it simply, see it in life, the real term. 117 terms in all.
- **Earned money:** ₹2,000 for finishing a lesson plus ₹1,000 per correct answer. A retry pays only the improvement, so guessing doesn't pay.
- **Golden assumption:** needs (food, home, fees, travel) are covered by family, so practice money is only for wants and savings, with a suggested 60/40 split.
- **Consequences:** 25 life events and scam alerts. Each choice is good, mixed or bad, moves money and meters, and explains why in one line. "What if" previews the other choices, and one try-again is allowed.
- **A portfolio that reacts:** wallet, savings, emergency fund, recurring and fixed deposits, gold, mutual fund, shares, things you own that lose value, and loans. A Money Health Score (0–100) names the next step to improve it. The credit score built from on-time payments sets the loan rate and limit.
- **Tools and Scam Shield:** 18 tools (SIP, EMI, FD, RD, inflation, tax slabs, Budget Builder, UPI Practice and more). Scam Shield mixes scams with genuine messages and teaches red flags and where to report (1930, cybercrime.gov.in).
- **Finale:** a 10-question final quiz, then an end-of-game summary with before and after, milestones and a money plan.
- **Two portals:** `/start` has two doors, Student and NGO. The NGO view (`/ngo`) is a class dashboard on a sample class of 31 learners plus this device as a live row: progress and first-try quiz scores by level, weak spots (the risky first choices in life events), scam safety, a nickname-only learner table with CSV export, and a privacy panel. No login in the demo.
- **Hindi and Telugu:** a header switch changes the whole app to Hindi or Telugu (about 99% of text translated). The English money term stays in brackets so words match what students see in banks and apps; translations are marked as drafts for native review.

3. Curriculum

Level	Theme	Lessons
1	Money basics	Income and needs vs wants · wants into goals · budget and pay yourself first · assets, liabilities, net worth
2	Banking and digital money	Bank accounts · cards, UPI and safe payments · how banks use your money
3	Safety net and saving habit	Emergency fund · insurance in plain words · recurring deposit
4	Borrowing and credit	Loans and EMIs · credit cards and buy-now-pay-later · credit score · formal vs informal borrowing
5	Making money grow	Inflation · simple vs compound interest · FD and small savings
6	Investing, one asset at a time	Risk and diversification · gold · mutual funds and SIP · shares · how options behaved over time
7	Taxes and the first paycheck	Why taxes and how slabs work · first paycheck · tax on savings and gains
8	Fraud-proof	Phishing and pressure tricks · investment and job scams (5-point check) · protect, report, recover
9	The big picture	Portfolio check-up, final quiz, money plan

4. How it is built

- **Content as data:** lessons, events and terms are JSON files checked by schemas. Every on-screen string is a translation key, so a new language is new files plus review, and the English term can be kept beside each translation.
- **Deterministic engine:** a pure TypeScript reducer over the game state. Month-close applies automatic payments, interest, depreciation, a fixed scripted market and credit rules. There is no randomness, so the same choices always give the same result and class results stay comparable.
- **One finance library:** SIP, lump sum, goal SIP, EMI, FD, RD, inflation, depreciation and slab tax feed both the engine and the calculators; every game number lives in one config file.
- **App:** React, Vite and TypeScript; mobile-first at 360 px and laptop-friendly; a plain-CSS whiteboard theme with self-hosted fonts. Progress is saved on the device; no account is needed.
- **Process:** parallel workstreams merged by pull request; every merge runs typecheck, all tests and a production build first.

5. Guardrails, safety and privacy

- **Educate, never recommend:** no product, fund, broker or app names. Comparisons are neutral (index vs active funds, direct vs regular plans). Investing lessons carry a "not advice" banner, and a lint test blocks advice words such as "best" or "you should buy".
- **Assumptions, not promises:** every projection is labelled as one, market scenarios are called pretend markets, and returns can be negative. Facts we could not verify on an official source (for example tax slabs or credit-score bands) are flagged and labelled approximate.
- **Privacy:** no sign-up, no names, phone numbers or emails, no analytics or trackers. Scam examples use no real brands or people.
- **Safe to fail:** balances never go below zero, a missed payment explains itself, and there is never a game over.

6. Testing and evidence

Area	Tests	What they prove
Content checks	433	Schemas, every text key exists in English, Hindi and Telugu, 5–8 cards and exactly 3 quiz questions per lesson, terms defined, rupee figures in lesson text equal the formula results, guardrail lint.
Engine	132	Month-close order, shortfall and credit rules, scripted market episodes, health-score bands, events, what-if leaves state untouched, deterministic replay.
Finance and core	46	Hand-computed values: SIP ₹10,000 × 15 years at 10% = ₹41.79 lakh; EMI on ₹6,000 over 6 months at 18% = ₹1,053; FD ₹6,000 for 6 months at 6.5% = ₹6,195; RD ₹500 × 6 ≈ ₹3,057.
Screens, tools, Scam Shield, NGO	123	Routes render (both portals); installable app manifest; language switch; lesson 1.1 played end to end; retry payout; final quiz; Plan the Month, events and summary; every calculator; Scam Shield scoring; NGO class figures.
Total	**734**	

Screens were checked by hand at phone width (360 px) and laptop width (1280 px) for layout, tap-target size and sideways scrolling. **Learning evidence** comes from first-attempt quiz scores per concept and the 10-question final quiz. With no pre-test baseline we report scores, not "improvement".

7. Fit with the success measures

Measure	How Capito meets it
Simple	One idea per card, plain words first, one decision per screen.
Relatable	Hyderabad situations: chai stalls, festival sales, phone EMIs, UPI, stipends; students add their own wants and goals.
Persona-led	The student's own choices, wants and goals shape their portfolio; the same events lead different students to different outcomes.
Language-ready	Built: English, Hindi and Telugu from a header switch, with the English term kept in brackets. Every string is a key, so another language is new files plus review.
Game-based	Practice money, portfolio, meters, milestones, consequences and a try-again.
Intuitive UI	Big buttons, a Continue button that always knows the next step, works on a phone and a laptop.
NGO-integrated	Built for the demo: an NGO portal with a class dashboard (progress, quiz scores, weak spots, scam safety, nickname-only table, CSV export) on a sample class. Real class codes need server sync (next steps).

8. Limitations and next steps

- **Not yet built:** real class codes with server sync (the NGO dashboard runs on a sample class today; the data layer has a storage interface ready for it) and offline mode.
- **Translations need native review:** Hindi and Telugu text is marked as a draft until native speakers check it.
- **Not yet tested with students:** next is a small pilot with Grade 11–12 learners to check reading level and first-attempt quiz scores.
- **Facts change:** tax slabs, interest rates and scheme rules live in one dated config and need a yearly review.

Practice money and pretend markets only. Capito explains how money works; it does not give financial advice or recommend any product.